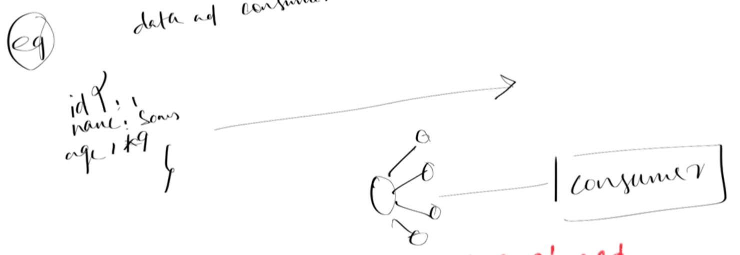
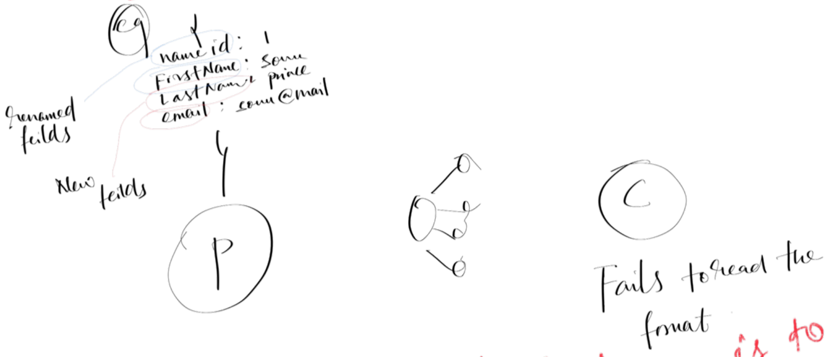
SpringBoot Kafka Avro Producer & Consumer | Schema Registry

* lets assume wehave a Producer & Consumer, we're Sending the data and consumer is Consuming it, and processing the data as Configured (pub & Sub mechanism)

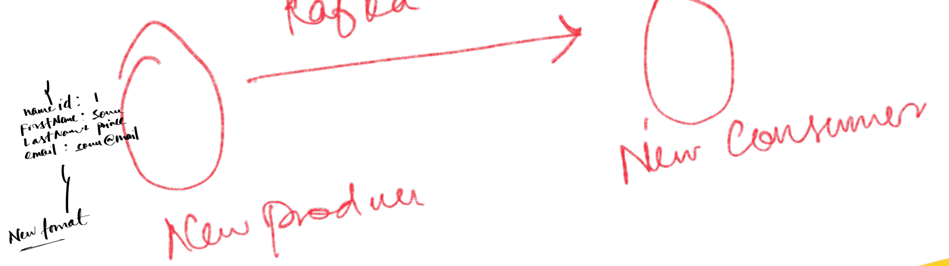

eg
{
id: 1
name: Sonu
age: 29
}
consumer

producer

Scenario: what if the dataformat is chnged.


eg
{
name id: 1
FirstName: Sonu
LastName: prince
email: sonu@mail
}
renamed feilds
New feilds
P
C
Fails to read the format

how to over come this issue is to create a New P & C and introduce the New format


{
name id: 1
FirstName: Sonu
LastName: prince
email: sonu@mail
}
New format
Kafka
New producer
New Consumer

This requires a too much work & is not a fesible Solution

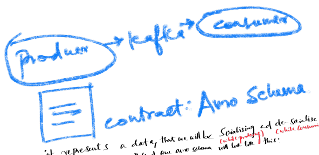
How to handle data changes & Schema evaluation → validating the data matching the format only available in confluent kafka

→ So thats where Confluent kafka introduced an concept called AVRO Schema.

To store the schema, they have a component called Schema registry.

Avro Schema

it's a Contract between Our Producer and Consumer.


Producer
kafka
consumer
contract: Avro Schema

* it represents a data, that we will be Serializing (while producing) and de-Serialise (while consuming).
* for a data which we will send, One avro schema will look like this:

```
{
 "id": 1,
 "name": "John",
 "email": "john@gmail.com"
 "DOB": "01/01/24"
}
```

Data

→

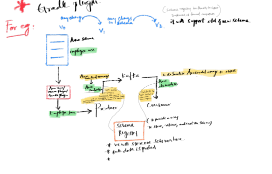
```
{
 "type": "record",
 "name": "Employee",
 "namespace": "com.ashfaq.dev",
 "fields": [
 { name: id,
   type: long
 },
 { name: Name,
   type: String
 },
 { name: email,
   type: String
 },
 { name: DOB,
   type: String
 }
 ]
}
```

Avro Schema / Contract

* Need to Create an Avro Schema where in Schema we will define the fields which Producer will Send and Consumer will Consume. in ~~Each~~ Case The producer is not Sending any of The fields or adding any new fields There should be any impact at consumer side, also any changes in Schema file it will a new Version.
* How to use The Contract / Schema post creating The scheme we need to generate the Object out of The Schema. we also need to define which are optional and mandatory fields in the file. Then we have to Generate the avro object. Converting the Avro Schema to avro object we can use

→ Converting
- Avro tool
- Maven plugin
- Gradle plugin.

For eg:


any change
V0
V1
any changes in Schema
V2
(Schema registry has flexibility to support backward and forward compatibility)
it will support old & new schema.
Avro Schema
employee.avsc
Avro tool/ Maven plugin/ Gradle plugin
Employee.java
Producer
Avro encoded message
Avro Serializer
Kafka
To deSerialize Avro encoded message to object
Avro deserializer
Consumer
Schema Registry
(To provide a way to store, retrieve, and evol the Schema)
* we will store one schema here
* once data is pushed
*

Getting Started with code

Spring Boot 3x

→ web, Spring for apache kafka.

→ Setting up Confluent Kafka using docker-compose file

Docker-compose

{
- → Zookeeper:
- → Broker: depends on
- → Kafka-tools: → Admin tools (optional)
- → Schema-registry → Store and Provide the Schema for validation.
- → Control-Center UI

Start the docker-compose yml file

docker-compose up -d

Moving to Readme.md